



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS

(First paper)

NQF LEVEL 2

(10501042)

11 November 2025 (Y-paper)

13:00–16:00

Non-programmable scientific calculators may be used.

This question paper consists of 7 pages, 1 formula sheet and 1 graph sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
MATHEMATICS
(First paper)
NQF LEVEL 2
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
 2. Read all the questions carefully.
 3. Number the answers according to the numbering system used in this question paper.
 4. Write your EXAMINATION NUMBER on the graph sheet in the space provided. Detach the sheet, place it in your ANSWER BOOK and hand it in together with the ANSWER BOOK.
 5. Use only a black or blue pen.
 6. Write neatly and legibly.
-

2)

-3-

QUESTION 1

- 1.1 Convert the following decimal fractions to the form $\frac{a}{b}$; where $a; b \in \mathbb{Z}$ and $b \neq 0$.
Express your answer in its simplest form.

1.1.1 0,015

1.1.2 5,16

(2 × 2) (4)

- 1.2 Simplify the following by using the laws of exponents: (Leave answers with positive exponents and in surd form where applicable).

1.2.1
$$\frac{\sqrt[3]{a^9 b^{\frac{3}{4}} c^3}}{(a^2 b^{-\frac{2}{3}} c^{-1})^3}$$
 (3)

1.2.2
$$\frac{(-5x^{-2}) \times ((y^3)(-3z^{-5}))^2}{((x^{-1})(y^4))^2}$$
 (4)

- 1.3 Simplify the following by using surd laws. Leave answers in surd form.

1.3.1
$$\frac{\sqrt{32} + 2\sqrt{18} - \sqrt{50}}{\sqrt{200} + \sqrt{72} - \sqrt{125}}$$
 (3)

1.3.2
$$\sqrt{\sqrt{54} \times \sqrt{150}} - \sqrt{(\sqrt{16} + \sqrt{36})}$$
 (3)

- 1.4 Simplify the following expression by rationalising the denominator:

$$\frac{12}{\sqrt{2} + \sqrt{3}}$$
 (3)

- 1.5 The formula to calculate the volume of a right cone is given by: $V = \frac{1}{3}\pi r^2 h$, with V = volume, h = height and r = radius.

1.5.1 Make 'r' the subject of the technical formula $V = \frac{1}{3}\pi r^2 h$. (3)

1.5.2 Calculate the value of 'r' if Volume is $6,514 m^3$ when the height is $3.123m$. (2)

- 1.6 The sum of the first 5 terms of an arithmetic progression is 90. If the first term is 2, calculate the common difference of the sequence. (3)

- 1.7 Calculate the 9th term of an arithmetic sequence if the first term is -4 and the constant difference is -5. (3)

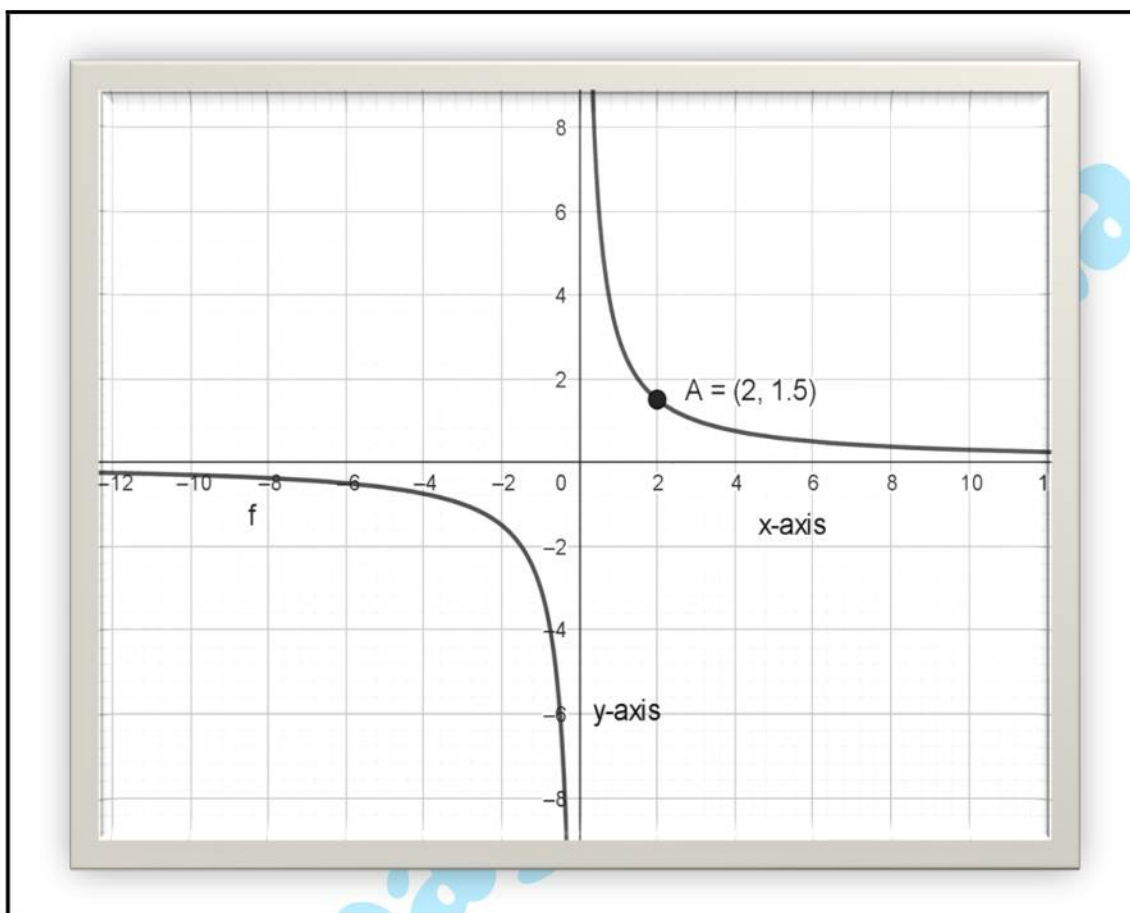
[30]

2)

-4-

QUESTION 2

2.1 Given below is the graph of f with point A (2, 1.5).



2.1.1 Give the name of the graph of f . (1)

2.1.2 Determine the equation of the graph of f . (3)

2.1.3 Is this graph of f a function or relation? (1)

2.1.4 Is this graph of f continuous or discontinuous? (1)

2.2 Given: $f(x) = -2x^2 + 8$

2.2.1 What is the name of the graph given in QUESTION 2.2 above? (1)

2.2.2 Calculate both the x-intercept and y-intercept of the graph in QUESTION 2.2 above. (3)

2.2.3 What is the axis of symmetry of the graph given in QUESTION 2.2 above? (2)

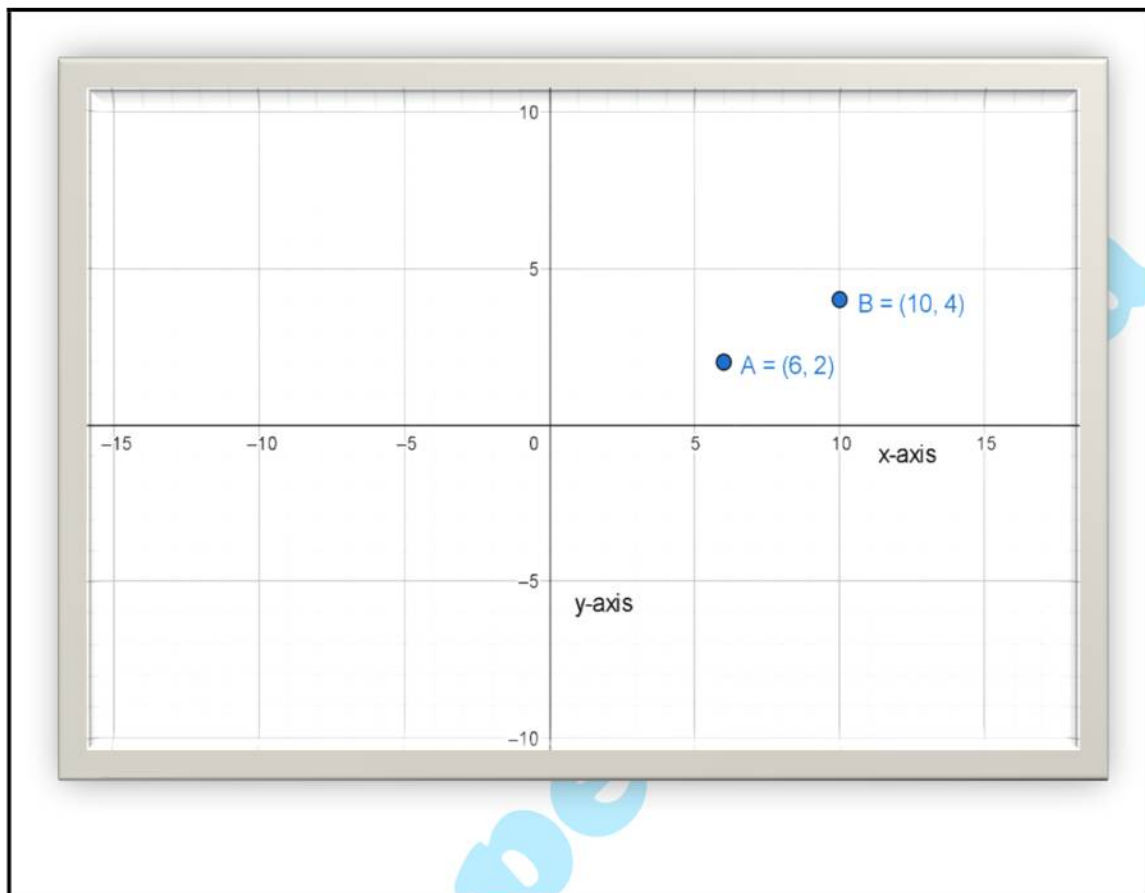
2.2.4 Draw the graph of $f(x)$ and show the x-intercept and y-intercept on it. (3)

2.)

-5-

2.3

Study the sketch given below:



- 2.3.1 Calculate the gradient of the line that passes through point A and point B as shown in the diagram above. (3)
- 2.3.2 Determine the equation of the line that passes through point A and point B as shown in the diagram above. (3)
- 2.3.3 Determine both the x-intercept and the y-intercept. (3)
- 2.3.4 Is the gradient of this line that passes through point A and point B, a negative or positive gradient? (1)
- [25]

2)

-6-

QUESTION 3

3.1 Factorise the following completely:

3.1.1 $5x^3y - 10x^2y^2 + 15xy^3$ (3)

3.1.2 $x^3 - 3x^2 - 4x + 12$ (3)

3.2 Simplify the following:

3.2.1 $\frac{(x^2 - 1)}{x + 1} - \frac{(x - 1)(x + 1)}{x + 2}$

3.2.2 $(x - \sqrt{2})(\sqrt{x} + 2x^2)$

3.2.3 $\frac{x^2 + x - 6}{x^2 + 7x + 12} \div \frac{x - 2}{x(x + 4)}$

(3 × 3) (9)

3.3 Solve for x:

3.3.1 $(3^x)(3^{x-2}) = 27$ (3)

3.3.2 $-2x^2 + x = -3x(-2x + 1)$ (3)

3.4 Solve algebraically for x and y in the following equations:

$2x + 3y + 4 = 0$

and

$x + 2y - 5 = 0$ (4)

[25]

QUESTION 4

4.1 Define the following financial terms in your own words and understanding:

4.1.1 Salary

4.1.2 Profit

4.1.3 Audit

(3 × 2) (6)

4.2 An amount of money totalling to R8245 is deposited into a savings account.

Calculate the final amount that will be available after three years if the interest is calculated at 7,5 % simple interest annually.

(5)

4.3 A farmer buys a tractor for R650 000 through a hire purchase agreement. The deal entails paying off the loan in five years through monthly payments with an annual interest rate of 13%.

Determine mathematically:

4.3.1 The total amount the farmer will pay for the tractor at the end of the purchase agreement.

4.3.2 The monthly repayments the farmer will make.

4.3.3 The total interest paid by the farmer at the end of the fifth year.

(3 × 3) (9)
[20]

TOTAL: 100

.2)

FORMULA SHEET

$$1. \quad a^m \times a^n = a^{m+n}$$

$$2. \quad a^m \div a^n = a^{m-n}$$

$$3. \quad (a^m)^n = a^{m \times n}$$

$$4. \quad (a^m b^n)^p = a^{mp} \cdot b^{np}$$

$$5. \quad \left(\frac{a^m}{b^n}\right)^p = \frac{a^{mp}}{b^{np}}$$

$$6. \quad a^{-n} = \frac{1}{a^n}$$

$$7. \quad a^0 = 1$$

$$8. \quad \sqrt[n]{a^m} = a^{\frac{m}{n}}$$

$$9. \quad T_n = a + (n - 1)d$$

$$10. \quad S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$11. \quad S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$12. \quad I = \frac{Prt}{100} \quad \text{or} \quad A_t = P(1 + in)$$

$$13. \quad A_t = Ao \left(1 + \frac{r}{100 \times m}\right)^{tm} \quad \text{or} \quad A_t = P(1 + i)^n$$

$$14. \quad i = \frac{r}{100}$$

2)

GRAFIEKPAPIER



DEPARTMENT OF EDUCATION
DEPARTEMENT VAN ONDERWYS

BOE 9/9

GRAPH PAPER • GRAFIEKPAPIER

(Return this sheet with the other answers)
(Lewer hierdie blad in saam met u antwoordsboek)

Examination number
Eksamennommer

